

MODULE 2 JOINTS

Articulations and Joints.

JOINTS

A joint is where two or more bones meet. This chapter covers both classifications, the three structural classes in detail, the synovial joint and its accessory structures, the movements, the shoulder and the knee, and the common disorders.

BY THE END

1. Classify joints by structure into fibrous, cartilaginous, and synovial, and by function into synarthrosis, amphiarthrosis, and diarthrosis.
2. Name the six types of synovial joint and give a body example of each.
3. Use the parts of a synovial joint to explain how it allows movement with little friction.
4. Name the movements that occur at synovial joints.
5. Describe the shoulder and the knee, and explain what makes each one vulnerable.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Joints

drsrennie-stack.github.io/new-build-bio4-solano/joints-concept-videos.html

Classifying Joints

Joints are classified two ways: by structure and by function. Learn both, because a tagged joint can be asked either way.

STRUCTURAL CLASSIFICATION: CAVITY PRESENT, AND WHAT HOLDS THE BONES

Structural class	Synovial cavity	Connective tissue holding the bones
Fibrous joint	None	Dense connective tissue rich in collagen fibers
Cartilaginous joint	None	Cartilage, hyaline or fibrocartilage
Synovial joint	Present	The dense irregular connective tissue of an articular capsule

FUNCTIONAL CLASSIFICATION: DEGREE OF MOVEMENT

Functional class	Degree of movement	Examples
Synarthrosis	Immovable	Sutures of the skull, gomphoses
Amphiarthrosis	Slightly movable	Syndesmoses, symphyses
Diarthrosis	Freely movable	All synovial joints

Fibrous Joints

No synovial cavity. The bones are held closely by dense connective tissue, and they permit little or no movement.

STRUCTURE, MOBILITY, EXAMPLES

Type	Structure	Mobility	Examples
Suture	Thin layers of dense irregular connective tissue, with interlocking edges that add strength and absorb shock	Immovable in adults, a synarthrosis. Slightly movable in infants and children, an amphiarthrosis	Only between the bones of the skull. The metopic suture is a frontal suture that persists past about age six
Syndesmosis	More dense connective tissue and more space than a suture, the bones united by a ligament	Slightly movable, an amphiarthrosis	The distal tibiofibular joint, united by the anterior tibiofibular ligament
Gomphosis	A cone-shaped peg held in a socket by the periodontal ligament	Immovable, a synarthrosis	A tooth in its bony socket
Interosseous membrane	A substantial sheet of dense connective tissue binding neighbouring long bones	Slightly movable, an amphiarthrosis	Between the radius and ulna, between the tibia and fibula
Synostosis	A joint where two separate bones fuse completely into one	Immovable	A suture that ossifies in adulthood

Cartilaginous Joints

No synovial cavity, and little or no movement.

THE TWO TYPES, BY CONNECTING CARTILAGE

Type	Connecting cartilage	Mobility	Examples
Synchondrosis	Hyaline cartilage	Immovable, a synarthrosis	The epiphyseal growth plate, which becomes a synostosis when bone replaces the cartilage. The joint between the first rib and the manubrium
Symphysis	A broad, flat disc of fibrocartilage	Slightly movable, an amphiarthrosis	The pubic symphysis, the manubriosternal joint, the intervertebral joints between vertebral bodies. All symphyses lie in the midline

Synovial Joint Structure

A synovial joint has a synovial cavity between the articulating bones and allows considerable movement. All synovial joints are diarthroses.

CARTILAGE AND CAPSULE

Structure	What it is
Articular cartilage	Hyaline cartilage covering the articulating bone surfaces. Smooth and slippery, it reduces friction and absorbs shock but does not bind the bones
Articular capsule	A sleeve-like capsule that surrounds the joint, encloses the synovial cavity, and unites the bones

Structure	What it is
Fibrous membrane	The outer layer of the capsule. Dense irregular connective tissue attached to the periosteum, strong and flexible
Ligaments	Thickened fibrous bundles of the fibrous membrane that resist strain and prevent dislocation
Synovial membrane	The inner layer of the capsule. Areolar connective tissue with elastic fibers
Articular fat pads	Accumulations of adipose tissue that may cushion the joint

SYNOVIAL FLUID, AND THE FOUR THINGS IT DOES

Feature	What it is or does
Synovial fluid	A viscous, clear, pale-yellow fluid of hyaluronic acid and interstitial fluid, secreted by the synovial membrane
Lubrication	Forms a thin film that reduces friction between the joint surfaces
Nourishment	Supplies oxygen and nutrients to, and removes wastes from, the avascular articular cartilage
Defence	Contains phagocytic cells that clear microbes and debris from normal wear
Warm-up effect	Gel-like at rest, it thins with movement, which is why warming up before exercise eases the joints

Accessory Structures of Synovial Joints

LIGAMENTS, DISCS, BURSAE, AND SUPPLY

Structure	What it is
Extracapsular ligaments	Ligaments that lie outside the articular capsule, as the collateral ligaments of the knee
Intracapsular ligaments	Ligaments inside the capsule but kept out of the synovial cavity by synovial folds, as the cruciate ligaments of the knee
Articular discs (menisci)	Crescent-shaped fibrocartilage pads between the bone surfaces. They improve fit, absorb shock, distribute weight, and spread synovial fluid
Labrum	A fibrocartilage rim around the socket of a ball-and-socket joint that deepens it. Found at the shoulder and hip
Bursae	Small sac-like structures filled with fluid, placed to reduce friction between skin, bone, muscle, tendon, or ligament
Tendon sheaths	Tube-shaped bursae that wrap a tendon where it passes through a tight space, as at the wrist and ankle
Nerve supply	The same nerves that supply the muscles moving the joint. They carry pain information and help adjust movement
Blood supply	Arteries near the joint branch into the capsule and ligaments, veins carry wastes away. The articular cartilage is fed instead by the synovial fluid

Movements at Synovial Joints

GLIDING AND ANGULAR MOVEMENTS

Movement	What it is
Gliding	Simple back-and-forth or side-to-side movement, no change in the angle between bones
Flexion	Decreases the angle between bones, usually in the sagittal plane
Extension	Increases the angle between bones, the reverse of flexion
Hyperextension	Extension that continues past the anatomical position
Lateral flexion	Bending the trunk or head sideways, in the frontal plane
Abduction	Movement away from the midline of the body
Adduction	Movement toward the midline of the body
Circumduction	Movement of the distal end of a part in a circle

ROTATION AND SPECIAL MOVEMENTS

Movement	What it is
Rotation	A bone turns around its own long axis. Medial rotation toward the midline, lateral rotation away

Movement	What it is
Elevation, depression	Lifting a body part upward, then lowering it
Protraction, retraction	Moving a body part forward, then back to position
Inversion, eversion	Turning the sole of the foot inward, then outward
Dorsiflexion, plantar flexion	Bending the foot up at the ankle, then pointing the toes down
Supination, pronation	Turning the palm to face anteriorly, then posteriorly
Opposition	Moving the thumb across the palm to touch the fingertips

The Six Types of Synovial Joints

Sorted by the shape of their surfaces, which sets the axes of movement. Uniaxial moves around one axis, biaxial around two, triaxial around three.

SHAPE, AXES, EXAMPLES			
Joint type	Shape of the surfaces	Axes of movement	Examples
Plane joint	Flat or slightly curved surfaces that glide	Biaxial or triaxial gliding	The intercarpal and intertarsal joints
Hinge joint	A convex surface fits a concave one	Uniaxial, flexion and extension	The elbow, ankle, and interphalangeal joints
Pivot joint	A rounded surface fits a ring of bone and ligament	Uniaxial, rotation	The atlantoaxial and radioulnar joints
Condyloid joint	An oval projection fits an oval depression	Biaxial	The radiocarpal and metacarpophalangeal joints
Saddle joint	Each surface is both concave and convex	Biaxial	The carpometacarpal joint of the thumb
Ball-and-socket joint	A ball fits a cup	Triaxial	The shoulder and hip joints

Factors Affecting Range of Motion

WHAT SETS THE LIMITS	
Factor	How it limits movement
Bone shape	The structure and fit of the articulating bones
Ligament tension	Taut ligaments restrict movement and direct the bones
Contact of soft parts	One body surface meeting another can limit movement
Hormones	Relaxin loosens the pubic symphysis and pelvic ligaments

The Shoulder Joint

The glenohumeral joint, a ball-and-socket joint between the head of the humerus and the glenoid cavity of the scapula. It is the most freely movable joint in the body, and the price of that mobility is instability.

STRUCTURE AND SUPPORTING LIGAMENTS

Structure	What it is
Type	A ball-and-socket joint, also called the glenohumeral or humeroscapular joint
Articulating bones	The head of the humerus and the glenoid cavity of the scapula
Why it moves so freely	The articular capsule is loose and the glenoid cavity is shallow relative to the large humeral head
Articular capsule	A thin, loose sac that envelops the joint, from the glenoid cavity to the anatomical neck of the humerus
Coracohumeral ligament	Runs from the coracoid process to the greater tubercle. Strengthens the superior part of the capsule
Glenohumeral ligaments	Three thickenings of the anterior capsule that stabilize the joint near the limits of motion
Transverse humeral ligament	A narrow band between the greater and lesser tubercles that holds the long head of the biceps tendon in place
Glenoid labrum	A rim of fibrocartilage around the glenoid cavity that slightly deepens and enlarges it
Bursae	Four are associated with the joint: the subscapular, subdeltoid, subacromial, and subcoracoid bursae
Movements	Flexion, extension, hyperextension, abduction, adduction, medial and lateral rotation, and circumduction
Source of stability	The ligaments add little strength. Most stability comes from the rotator cuff muscles
Rotator cuff	The supraspinatus, infraspinatus, teres minor, and subscapularis, which hold the humeral head in the glenoid cavity

MNEMONIC

The rotator cuff is SITS: Supraspinatus, Infraspinatus, Teres minor, Subscapularis.

The Knee Joint

The tibiofemoral joint, the largest and most complex joint in the body. A modified hinge joint, and the joint most often injured.

STRUCTURE, LIGAMENTS, AND MENISCI

Structure	What it is
Type	A modified hinge joint, the largest and most complex joint in the body
Three joints, one cavity	A medial and a lateral tibiofemoral joint between the femoral and tibial condyles, and a patellofemoral joint between the patella and the femur

Structure	What it is
Articular capsule	The capsule that encloses the joint
Patellar retinacula	The medial and lateral retinacula, fused quadriceps tendon and fascia, strengthen the anterior surface
Patellar ligament	The continuation of the quadriceps tendon from the patella to the tibial tuberosity. Strengthens the anterior surface
Oblique popliteal ligament	A broad ligament that strengthens the posterior surface of the joint
Arcuate popliteal ligament	Strengthens the lower lateral part of the posterior surface
Anterior cruciate ligament	An intracapsular ligament, crossing within the joint. Stops the tibia from sliding forward. Torn in roughly 70 percent of serious knee injuries
Posterior cruciate ligament	An intracapsular ligament, crossing within the joint. Stops the tibia from sliding backward. Important on stairs and inclines
Medial meniscus	The C-shaped fibrocartilage disc on the medial side of the joint
Lateral meniscus	The more circular, O-shaped disc on the lateral side
Transverse ligament	The band that joins the two menisci to each other

Knee injuries

WHAT GOES WRONG AND WHAT IT MEANS

Injury	What it is
Unhappy triad	A lateral blow to a planted knee that tears three structures at once: the tibial collateral ligament, the medial meniscus, and the anterior cruciate ligament
ACL and women	The anterior cruciate ligament is torn three to six times more often in women, from a narrower intercondylar space, a wider pelvis and greater knee angle, ligament-loosening hormones, and less protective muscle strength
Swollen knee	Immediate swelling is blood from torn vessels. Delayed swelling is excess synovial fluid
Dislocated knee	Displacement of the tibia relative to the femur, most often anterior. It can damage the popliteal artery
PRICE	The first-aid approach to a knee injury: protection, rest, ice, compression, elevation

Common Joint Disorders

DISORDER AND DESCRIPTION

Disorder	What it is
Double-jointed	Greater ligament flexibility and range of motion. The joints are less stable and dislocate more easily

Disorder	What it is
Bursitis	Inflammation of a bursa, from repeated overuse, infection, or rheumatoid arthritis, with pain, swelling, and limited movement
Torn meniscus	A tear of the knee cartilage, common in athletes, repaired by arthroscopy with a fibre-optic camera
Rheumatism	Any painful disorder of the supporting structures, bones, ligaments, tendons, or muscles, not caused by infection or injury
Arthritis	A form of rheumatism with swollen, stiff, painful joints. The leading cause of physical disability in adults over 65
Osteoarthritis	A progressive, degenerative loss of joint cartilage from aging and wear, in the large weight-bearing joints
Rheumatoid arthritis	An autoimmune disease that attacks the cartilage and joint linings, usually on both sides, inflaming the synovial membrane
Gouty arthritis	Sodium urate crystals deposit in joints, often the base of the big toe, eroding the cartilage with inflammation and pain

Classification: joints by structure

THE STRUCTURAL CLASSES AND THEIR SUBTYPES

Class	Subtypes and examples
Fibrous, joined by dense connective tissue	Sutures (skull), gomphoses (tooth sockets), syndesmoses (radius to ulna)
Cartilaginous, joined by cartilage	Synchondroses (epiphyseal plate, first rib), symphyses (pubic symphysis, intervertebral discs)
Synovial, joined across a fluid-filled cavity	Plane, hinge, pivot, condyloid, saddle, ball-and-socket

FREEDOM OF MOVEMENT HAS A COST

Synovial joints are freely movable, and that is exactly why they are the joints that get injured. A sprain is a stretched or torn ligament, the band that stabilizes a joint. Two arthritis types are worth telling apart: osteoarthritis strikes the large joints and is wear of the cartilage, while rheumatoid arthritis attacks the small joints first and actively destroys the cartilage and synovial membrane. Knowing the joint type tells you the movements to expect, and the ways the joint can fail.